Assignment 3

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Name : Ravish Ranjan
Course : MCA
Semester : 2nd semester
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Task 1: Finding the Full Email Header

Before analysis can begin, locate the full email header in your email client (Gmail/Outlook). The method differs depending on the platform.

Gmail

1. Open Gmail in a web browser and open any email (use a test/sample email provided by your instructor).
2. Click the three dots (More options) in the top-right corner of the email.
3. Select "Show original" from the menu. This will open a new window displaying the full raw email headers.
4. Copy the entire header text from the box shown on that page.

```
Received: from SA3PR19MB7370.namprd19.prod.outlook.com (::1) by
 MN0PR19MB6312.namprd19.prod.outlook.com with HTTPS; Tue, 19 Sep 2023
18:36:46
 +0000
Received: from BN0PR03CA0023.namprd03.prod.outlook.com
(2603:10b6:408:e6::28)
 by SA3PR19MB7370.namprd19.prod.outlook.com (2603:10b6:806:317::17) with
 Microsoft SMTP Server (version=TLS1_2,
 cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6792.27; Tue, 19
Sep
 2023 18:36:45 +0000
Received: from BN8NAM11FT066.eop-nam11.prod.protection.outlook.com
 (2603:10b6:408:e6:cafe::23) by BN0PR03CA0023.outlook.office365.com
 (2603:10b6:408:e6::28) with Microsoft SMTP Server (version=TLS1_2,
 cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6792.28 via
Frontend
 Transport; Tue, 19 Sep 2023 18:36:45 +0000
Authentication-Results: spf=temperror (sender IP is 137.184.34.4)
 smtp.mailfrom=ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06; dkim=none (message
not
 signed) header.d=none;dmarc=temperror action=none
 header.from=atendimento.com.br;compauth=fail reason=001
Received-SPF: TempError (protection.outlook.com: error in processing during
 lookup of ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06: DNS Timeout)
Received: from ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 (137.184.34.4) by
 BN8NAM11FT066.mail.protection.outlook.com (10.13.177.138) with Microsoft
SMTP
 Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id
```

```
 15.20.6813.19 via Frontend Transport; Tue, 19 Sep 2023 18:36:44 +0000
X-IncomingTopHeaderMarker:

OriginalChecksum:3B61F64750F88C5569DF38A496B2374685F23D8BC662A6A19B6823B2F6
745D54;UpperCasedChecksum:62071BC7A7CF5B0844A7B406B0E9EFCDAA2CB94988E687CF8
C56555AD4B52D30;SizeAsReceived:544;Count:9
Received: by ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 (Postfix, from userid 0)
    id 39DEA3F725; Tue, 19 Sep 2023 18:35:49 +0000 (UTC)
Content-type: text/html; charset=UTF-8
Content-Transfer-Encoding: base64
Subject: CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos
LIVELO expirando hoje!
From: BANCO DO BRADESCO LIVELO<banco.bradesco@atendimento.com.br>
To: phishing@pot
Message-Id: <20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-
06>
Date: Tue, 19 Sep 2023 18:35:49 +0000 (UTC)
X-IncomingHeaderCount: 9
Return-Path: root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
X-MS-Exchange-Organization-ExpirationStartTime: 19 Sep 2023 18:36:44.2236
 (UTC)
X-MS-Exchange-Organization-ExpirationStartTimeReason: OriginalSubmit
X-MS-Exchange-Organization-ExpirationInterval: 1:00:00:00.0000000
X-MS-Exchange-Organization-ExpirationIntervalReason: OriginalSubmit
X-MS-Exchange-Organization-Network-Message-Id:
 b9106deb-bd54-4815-e5c9-08dbb93f5fab
X-EOPAttributedMessage: 0
X-EOPTenantAttributedMessage: 84df9e7f-e9f6-40af-b435-aaaaaaaaaaaa:0
X-MS-Exchange-Organization-MessageDirectionality: Incoming
X-MS-PublicTrafficType: Email
X-MS-TrafficTypeDiagnostic:
 BN8NAM11FT066:EE_|SA3PR19MB7370:EE_|MN0PR19MB6312:EE_
X-MS-Exchange-Organization-AuthSource:
 BN8NAM11FT066.eop-nam11.prod.protection.outlook.com
X-MS-Exchange-Organization-AuthAs: Anonymous
X-MS-UserLastLogonTime: 9/19/2023 6:25:15 PM
X-MS-Office365-Filtering-Correlation-Id: b9106deb-bd54-4815-e5c9-
08dbb93f5fab
X-MS-Exchange-EOPDirect: true
X-Sender-IP: 137.184.34.4
X-SID-PRA: BANCO.BRADESCO@ATENDIMENTO.COM.BR
X-SID-Result: NONE
X-MS-Exchange-Organization-PCL: 2
X-MS-Exchange-Organization-SCL: 5
X-Microsoft-Antispam: BCL:9;
X-MS-Exchange-CrossTenant-OriginalArrivalTime: 19 Sep 2023 18:36:44.1298
 (UTC)
X-MS-Exchange-CrossTenant-Network-Message-Id: b9106deb-bd54-4815-e5c9-
08dbb93f5fab
X-MS-Exchange-CrossTenant-Id: 84df9e7f-e9f6-40af-b435-aaaaaaaaaaaa
X-MS-Exchange-CrossTenant-AuthSource:
 BN8NAM11FT066.eop-nam11.prod.protection.outlook.com
X-MS-Exchange-CrossTenant-AuthAs: Anonymous
X-MS-Exchange-CrossTenant-FromEntityHeader: Internet
```

```
X-MS-Exchange-CrossTenant-RMS-PersistedConsumerOrg:
 00000000-0000-0000-0000-000000000000
X-MS-Exchange-Transport-CrossTenantHeadersStamped: SA3PR19MB7370
X-MS-Exchange-Transport-EndToEndLatency: 00:00:02.6179349
X-MS-Exchange-Processed-By-BccFoldering: 15.20.6792.025
X-Microsoft-Antispam-Mailbox-Delivery:

wl:1;pcwl:1;ucf:0;jmr:0;ex:0;psp:0;auth:0;dest:I;OFR:TrustedSenderList;ENG:
(5062000305)(920221119095)(90000117)(920221120095)(91040095)(9050020)
(9075021)(9100341)(944500132)(2008001134)(4810010)(4910033)(9610028)
(9560006)(10180021)(9439006)(9310011)(9220031)(120001);
X-Message-Info:

qZelhIiYnPlgo3oeAkqKQrb/Je8fpvpPmRGjYwLej8PYXc5p/l16IG5I8gDUPoij+JWSvja0BAM
LtkgrOcbx5zEN7V98T2UZUZs4k8BX/DcDfI7QJ0t2aouiqx4ENvkR1M3sDKP/XN09+50x9Rxi6o
nUtDV4eqq36VUi2qAa0zCzkJwjdl3Y9DzNE1OkaWjrHAizeUyMZ/UtK/Pz9zhA2A==
X-Message-Delivery: Vj0xLjE7dXM9MDtsPTA7YT0wO0Q9MTtHRD0yO1NDTD0tMQ==
X-Microsoft-Antispam-Message-Info:
    =?utf-8?B?QTlXRFVaTVRhbmFzVTRkbVBTSFRSUURrQTRyaDhzZVczY2RROWF3bVVDTWdk?
=
 =?utf-8?B?bVU0VHJ2UU9wWUFLbXlFRWVUcmx1Z244ajk4M0JMRVYzZW9WVkE3NVZpK0dp?=
 =?utf-8?B?STVZSUFyRzdvQWNJeXEyNlNrZnBxcG9rZk5zQTAvMzBPbExJWWg2SFhEQWVv?=
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 =?utf-8?B?WFp2amorZ2JLVTd1UUVSb1R4bVg0czI2TUpyRE5HREQzQ0FrUldqK1BiUlJs?=
 =?utf-8?B?Y1hqT21UV2dJVmd5ZG9xVDk3U1BUZ0VvckVxM2tyS1BmRTRBPT0=?=
MIME-Version: 1.0
```
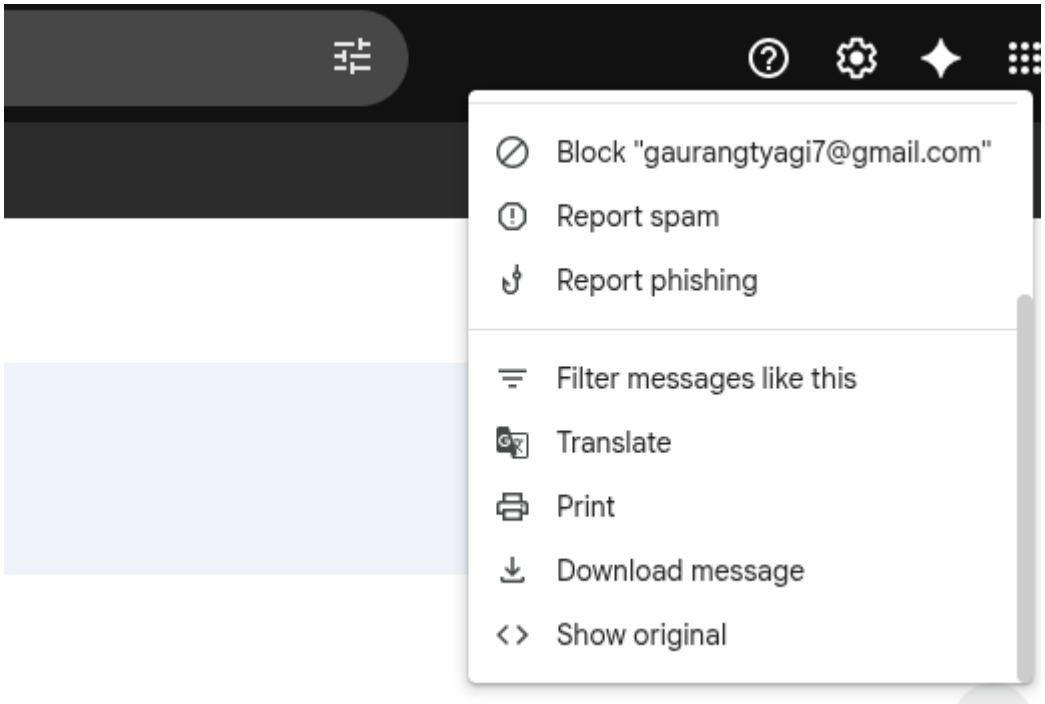


5. Paste a screenshot of the Gmail menu and the raw header window.



Download Original

Copy to clipboard


```
Delivered-To: ravishranjan2003@gmail.com
Received: by 2002:a17:907:a42:b0:b8f:9296:b369 with SMTP id be2csp1179547ejc;
        Sun, 1 Mar 2026 06:58:54 -0800 (PST)
X-Received: by 2002:a05:6a20:d498:b0:394:625f:c565 with SMTP id adf61e73a8af0-395c3a7faafmr9239485637.30.1772377134568;
        Sun, 01 Mar 2026 06:58:54 -0800 (PST)
ARC-Seal: i=1; a=rsa-sha256; t=1772377134; cv=none;
        d=google.com; s=arc-20240605;
        b=D/+g9bS3MrQINDwiLKGd3QWdNqdnmXt8aysvHl3AhcAFsJ8qaHAvrLrKc0KrvCpu46
         47mwanlB0AORNuJj+X5RFOuQo62pU6ZJvZpfAgqLc1mE7uNJfzFcFR8TvK5nhVENMVDQ
         ejbvQe7uT4moU9gcvmzE/B6p31j9/9dT6XqAG44EmR6qV70GxonEMlaRpSSqVg36H5hs
         lswBRavbnfhsOKH2E35ZVvKNnWwOpFXKHycze8ryS+I03cmZmm22wcE1Yrh3YtpLUyzy
         UayRpfxArpyOgZd/35efi2PgAvYxWP4cjQ6O44a3URt1F5RBbaL7uAnkX7mcCSIVUdq5
         h8Ig==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605;
        h=unique-id:feedback-id:accept:format:charset:mime-version:subject:to
         :reply-to:sender:from:message-id:date:dkim-signature;
        bh=T5gxnGT/rw8ukOXHTdo+JBEI5206csZamy4E5qdXXQo=;
        fh=1LgzGTEIIlCCIVHfpTbsZX0XbOJDhkF/Bqpo1TDn2lc=;
        b=QUcn6nmfyaMtueV2zPK0xkOWUkLsO/SngbiNMVPdGpp1CepfU1g0cRHv9lCKkAr4tB
         KNaMSY9KY19NbuPXn5NQerbLhBjYjq2t2O3hq8VfHHKw76kWEp/pZeIw08b6yJB0YJ4o
         f3y987sC6Ut9FCzKvaXzNeZV1JH0NYQfPx+TQTo+KowXKb/bfqFrdKSsTq1c7mRmJfI5
         +UmFz3STPilRa+hyHHzyj3xHV0S3hpL2NBe6CFvmZ/NJ4p+DFEbM4b7VTSH5HelU/SzW
         oAcThKQF0OhMmLt2dfS8CL9zxKiW91jevaPh9CBlL9PdyJ9xegNSZUpz5dS1lj+s9jiM
         3J0A==;
        dara=google.com
ARC-Authentication-Results: i=1; mx.google.com;
       dkim=pass header.i=@naukri.com header.s=2015 header.b=J4NmQb0W;
```

Task 2: Analyse the email in the EML Analyser

1. Load the provided .eml into an EML Analyser.
2. Record the top-level metadata:

Field	Value
Subject	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
From	banco.bradesco@atendimento.com.br
To	phishing@pot
Date	2023-09-19T18:35:49Z
Message-ID	20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
Return-Path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
MIME type	mime-version 1.0 content-type text/html; charset="UTF-8"
Body encoding	base64

3. List every MIME part, attachment, embedded object or referenced resource.

```
MIME part : text/html; charset="UTF-8"
attachments : None
embedded object : google font stylesheet,images
referenced source : html link tag
    <link href="https://fonts.googleapis.com/css2?
family=Signika:wght@300;500;700&amp;display=swap" rel="stylesheet">

    header.png
```

```
icone-superior.png
icone-rodape.png
```

4. Extract all URLs, domains, email addresses and any observable filenames URLs

```
https://blog1seguimentmydomaine2bra.me/
```

Domains

```
namprd19.prod.outlook.com
namprd03.prod.outlook.com
protection.outlook.com
outlook.office365.com
atendimento.com.br
```

Emails

```
banco.bradesco@atendimento.com.br
root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
phishing@pot
```

Filenames

```
header.png
icone-superior.png
icone-rodape.png
```

Submit:

5. Screenshot of email header analysis result in a tool

Verdicts

oleid N/A

There is no suspicious OLE file in attachments. N/A

Headers

Hops

Hop	From	By	With	Date (UTC)	Delay
1				2023-09-19T18:35:49Z	N/A
2	137.184.34.4	10.13.177.138, bn8nam11ft066.mail.protection.outlook.com	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.6813.19 via frontend transport	2023-09-19T18:36:44Z	a minute
3	bn8nam11ft066.eop-nam11.prod.protection.outlook.com, 2603:10b6:408:e6:cafe::23	bn0pr03ca0023.outlook.office365.com, 2603:10b6:408:e6::28	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.6792.28 via frontend transport	2023-09-19T18:36:45Z	a few seconds
4	2603:10b6:408:e6::28, bn0pr03ca0023.namprd03.prod.outlook.com	2603:10b6:806:317::17, sa3pr19mb7370.namprd19.prod.outlook.com	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.6792.27	2023-09-19T18:36:45Z	N/A
5	sa3pr19mb7370.namprd19.prod.outlook.com, ::1	mn0pr19mb6312.namprd19.prod.outlook.com	https	2023-09-19T18:36:46Z	a few seconds

Basic headers

Message ID	<20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06>
Subject	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
Date (UTC)	2023-09-19T18:35:49Z
From	banco.bradesco@atendimento.com.br
To	phishing@pot

Security headers

authentication-results	spf=temperror (sender IP is 137.184.34.4) smtp.mailfrom=ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06; dkim=none (message not signed) header.d=none;dmarc=temperror action=none header.from=atendimento.com.br;compauth=fail reason=001

X headers

x-ms-exchange-organization-expirationintervalreason	OriginalSubmit
x-microsoft-antispam	BCL:9;
x-incomingheadercount	9
x-sid-result	NONE
x-eopattributedmessage	0
x-microsoft-antispam-mailbox-delivery	wl:1;pcwl:1;ucf:0;jmr:0;ex:0;psp:0;auth:0;dest:I;OFR:TrustedSenderList;ENG:(5062000305)(920221119095)(90000117)(920221120095)(91040095)(9050020)(9075021)(9100341)(944500132)(2008001134)(4810010)(4910033)(9610028)(9560006)(10180021)(9439006)(9310011)(9220031)(120001);
x-ms-exchange-crosstenant-originalarrivaltime	19 Sep 2023 18:36:44.1298 (UTC)
x-ms-exchange-crosstenant-fromentityheader	Internet
x-incomingtopheadermarker	OriginalChecksum:3B61F64750F88C5569DF38A496B2374685F23D8BC662A6A19B6823B2F6745D54;UpperCasedChecksum:62071BC7A7CF5B0844A7B406B0E9EFCDAA2CB94988E687CF8C56555AD4B52D30;SizeAsReceived:544;Count:9
x-ms-exchange-organization-scl	5
x-ms-exchange-organization-authas	Anonymous
x-ms-exchange-eopdirect	true
x-sid-pra	BANCO.BRADESCO@ATENDIMENTO.COM.BR
x-ms-exchange-crosstenant-authsource	BN8NAM11FT066.eop-nam11.prod.protection.outlook.com
x-ms-exchange-crosstenant-rms-persistedconsumerorg	00000000-0000-0000-0000-000000000000
x-ms-publictrafficctype	Email
x-ms-exchange-crosstenant-network-messa	b9106deb-bd54-4815-e5c9-08dbb93f5fab

ge-id	
x-ms-exchange-crosstenant-authas	Anonymous
x-message-info	qZelhliYnPlgo3oeAkqKQrb/Je8fpvpPmRGjYwLej8PYXc5p/I16lG5l8gDUPoij+JWSvja0BAMLtkgrOcbx5zEN7V98T2UZUZs4k8BX/DcDfl7QJ0t2aouiqx4ENvkR1M3sDKP/XN09+50x9Rxi6onUtDV4eqq36VUi2qAa0zCzkJwjdl3Y9DzNE1OkaWjrHAizeUyMZ/UtK/Pz9zhA2A==
x-ms-traffictypediagnostic	BN8NAM11FT066:EE_|SA3PR19MB7370:EE_|MN0PR19MB6312:EE_
x-eoptenantattributedmessage	84df9e7f-e9f6-40af-b435-aaaaaaaaaaaa:0
x-ms-exchange-organization-messagedirectionality	Incoming
x-ms-exchange-transport-endtoendlatency	00:00:02.6179349
x-ms-exchange-organization-pcl	2
x-ms-exchange-processed-by-bccfoldering	15.20.6792.025
x-ms-exchange-organization-network-message-id	b9106deb-bd54-4815-e5c9-08dbb93f5fab
x-microsoft-antispam-message-info	A9WDUZMTanasU4dmPSHTRQDkA4rh8seW3cdQ9awmUCMgdmU4TrvQOpYAKmyEEeTrlugn8j983BLEV3eoVVA75Vi+Gil5YIArG7oAclyq26Skfpq pokfNsA0/30OlLlYh6HXDAeoDMBxHn30zgHdQgh44V7E4cRGr9q8dLQ4U8PGGTQLYg50SC79Kla8rbdNJMaXxiDJrrcJ1z/BeQQB+Dit+OI78ZgadlrD28l10GjfS5Ri6NGzhxMSrB9bHRbeOlp7c2DkcmJ4HZGQUqRx5UnkVGt+rrl+yVEd83aGnQl0pQt+bO5diP8HlWnrGkd4/gzGwWSZ7wR00t3k5yn1o4pzVb/KkcPUTPGHVk+aBqJHlqdDmSMXdEHfqkbHjxjaVufSozOANeFFr/WlYeJBqxOA1oretAr7MRpqYeHl2zQjoZ0QK4aUUHlLHX8SCuCUwVXPAYARZ7ehlgpvrEcqHB3v8iH98idTMNMaAnkPycmUwT2Oa+4TWvtLLrxGgiuQHKRnDl/CNpXk87PO7oj8dt3KNQ9zdqhkPguxYxBXS2T0NJwDVjeTimJJfzhdiQPd1h2ULYR6fZ5JuD4HbdHOHwDrQ+VA2DLs0YnEowS+TDpK+Q19PZXhcDzWbjDWKC6KoCly2BDxfFZiWACrT4r4ZCSExfV9ZNCkt9+g3qTdUfm+yVMdzqaBUKoF0NXE8s37xrKcLIR5xxozjn8VFfVz/ykAO4a5C1R7boM4WkzjjtUNRdc1IqjAMuiWtP1yxbtEGc1Y6ocVgJ56fCN+E7j3DjCL85fhLo0Uisy5I4DfshNBMrb4cG0qxluGrm54GnbBKz2+h2tjjxO1T6VxtWwEx734hUYaTehqClVd9esCMRh6EIB+h3yBEMqMibgeVmunNeTz0OTVrPlUeMGZPjP2Ju0e3VBhvW4sUmKH8Yp+jLV+QB63VYMdZmxgTw61gWIHuUeBr5w2Z1y6gMH/8xX/Fm2oSYo/gpD5/j+0whrdtAleRL4AKqGheJPdmJJkcmvrYy83Tzs9pXLGC67d+Uk33ZHsLSYqTTYFkzbPPmr696cX337lvP/L0BB70NxDMz3S2XkazqLDdTERyWO8b03bFI6ZTEhaz+J5gGv+CxrQXljbUGMHoadtW0O9rmeMLJ2zT3dgpUWO75PyudRZcEKNnS55Kb3vn0TFW8ZHAhQD55vpEfvQkbGlKm9liRJ/v/3vL4H+4wPr2HwnWnMEZLaw4ZB7nrJSsa7aYfQWstI90LrUb+Y23mo/zXyTxisERzMbXogysUqw+QLSTzfdGXYG7TgTVv7HehGpmo/yVVNaxvPtBIZk3xU7t9VdYMomn0cWeVKSa5hW40hm74NRe3SnO5j2G5Z2FTP4hNeUzivl132R11hzjcSZMTwHDa/mH6FMf0YP3Ba5xxrtCEDLdWRMcEXbxViPyJc7W9eYPUOoobVkrzRHZiErGB3teaKDKKDd2j3b5KkrSHLa+WaYSqDUfwTm0Q405yH4WASCtjqLYUmvqs2LUnapA4HBj8zZEB4q1yPrPNoUaPZHJ7h+bgYq67+uVhz7Jlcsn/8fp750cQUA6DtGWGDfbrIMKscV1iOPAGgzinwkLiOg9LwtTbqnqNiscux4RgIDHPpef5JcnTNyxjMFQ/3htymlYdCV4bo6XFsmqG9kxbweipkP7TyQcOCHqkDwRmTwFekquWIbly88wSVWy7ucfOUsOVP7ft25t/IX6/UPZ7XL59I2p/92LDGtvwhE3eF+9PjgSc3Jxh37dzLkcXCyANBWg1Pbk/RYECWdVlcLAYR75ECIj16y5a/iRs2I4IHGNBnWessjnEYj3lHjFxBM/amYWkCs6PkAW7S2RwOSig1gCqsKR8rHyA/LR5qjdfCG7rlHd2rKeKwNnVfiBf7e/F6Agksjxb VPWD6MoCy2HKpXGotO2+YmcVwTx5SLpEmbBUwLh1ZWdWKg6K40bfrrel17Om/2I5YsQvajOQh4LxpjU3ZzzKDI94zDc8WyPKMzbmTska14Rfyz/20CztdptNubn+CMuUeezqC/Cm0NY14Zz1hcZhWN9XT6AIVcwbc3dNoEh07Bm9c29BImiVs074GXQhKS89yF9wCwh6A2n3KP5qghQB+5BXzXWYQjHSuHLbbf2Aykr6KUggjkESlgT8imYdzvF++3VYCZgVaj1uyhKsXuylbVFq6GcDjw9LYC5RoqrsEXQCHwSqXDCHKfDsPeB4teJsEcPlVUDNiQxWOp5VXQ98wJgWzc/ZMD/Bi/T/fWy9PcxTDrk/DP5G2RG60RJre67BtnsAKpAf+DhkQT041hdDHCo9X5/40KUCBI60HTbo3S252o3tMlzG3bePqDYti4Lcsjw6Fh70h7Us0mwmaljJ7wT9hyxkoy3DOV6WegpGQrAxmu48Qv+Ewni85jh13/6toBfvNFle8LKPJSWFMOofrDYR9w50DRCnhB/jAJbX3lF5oU45xPbBCKgLobc7kovAV9sSOKUIqKwJhbbEYq1q0OtA+2rNRIIVoNofliELUgqU0dtLOvHd1MrhRiLykB27zX254Yf3XZvjj+gbKU7uQERoTxmX4s26MJrDNGDD3CAkRWj+PbRRIcXjOmTWglVgydoqT97SPTgEorEq3krKPfE4A==

Other headers

content-transfer-encoding	base64
received-spf	TempError (protection.outlook.com: error in processing during lookup of ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06: DNS Timeout)
return-path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
mime-version	1.0
content-type	text/html; charset="UTF-8"

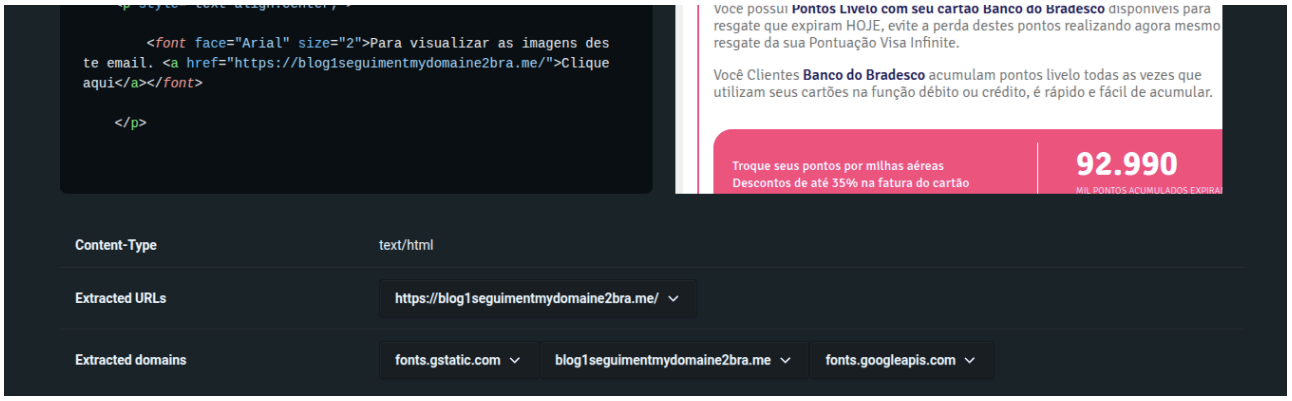
Bodies

text/html

```
<!DOCTYPE html><html lang="en"><head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-
8"><body style="background-color:rgb(241, 241, 241);">
```

Para visualizar as imagens deste email. Clique aqui

Banco do Bradesco (Livelo).



6. Results of Steps 2, 3 and 4.

Task 3: Header analysis with MX Toolbox

Paste the raw header into the MX toolbox Email header analyser and submit the following:

- Make a table and record the following: Analysis url

Field	Record
Displayed from the address	BANCO DO BRADESCO LIVELO banco.bradesco@atendimento.com.br
Return-path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
Message-ID	20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
Sender/originating IP	137.184.34.4
Received mail servers	ubuntu-s-1vcpu -> Outlook Protection -> Office365 internal servers
SPF result	temperror
DKIM result	none
DMARC result	temperror (no DMARC record)

1. Is there any mismatch between the displayed sender identity and the actual sender?

```
Yes
    displayed name : BANCO DO BRADESCO LIVELO
<banco.bradesco@atendimento.com.br>
    actual sender : root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 (random
ubuntu vps)
```

2. State whether the relay path is consistent with the claimed sender brand.

```
Not consistent
    the first hop is not a bank mail server but is a generic digital ocean
vps,so we can conclude that it is not a legitimate email from bank
```

3. From the MX Toolbox results in Task 3, note whether the SPF, DKIM, and DMARC checks passed or failed.

```
None passes properly
    SPF : temperror
    DKIM : none
    DMARC : temperror
```

4. Interpret the SPF result and explain what a temperror means operationally.

```
SPF result (from header) : spf=temperror (DNS Timeout)
    spf checkup failed due to dns lookup failure so the reciever couldn't
verify sender authorization

    this could be because of
        Misconfigured domain
        Non-existent domain
        Intentional evasion
```

5. Explain the significance of dkim=none for a bank-branded message.

```
dkmi=none means following things
    there is no cryptographic signature
    there is no verification of domain authenticity
    message integrity is not checked

these are necessary requirements for a banking mail
```

6. Based on your findings, explain below what these results tell you about the legitimacy of the email.

```
the mail is classified as non-legitimate for following reasons
    fake branding
    return path mismatch
    origin from a vps
    SPF failed
    DKIM is missing
    DMARC is missing
    microsoft has marked compauth=fail and SCL=5
```

Task 4: Tracing the Originating IP Address and analysing its reputation

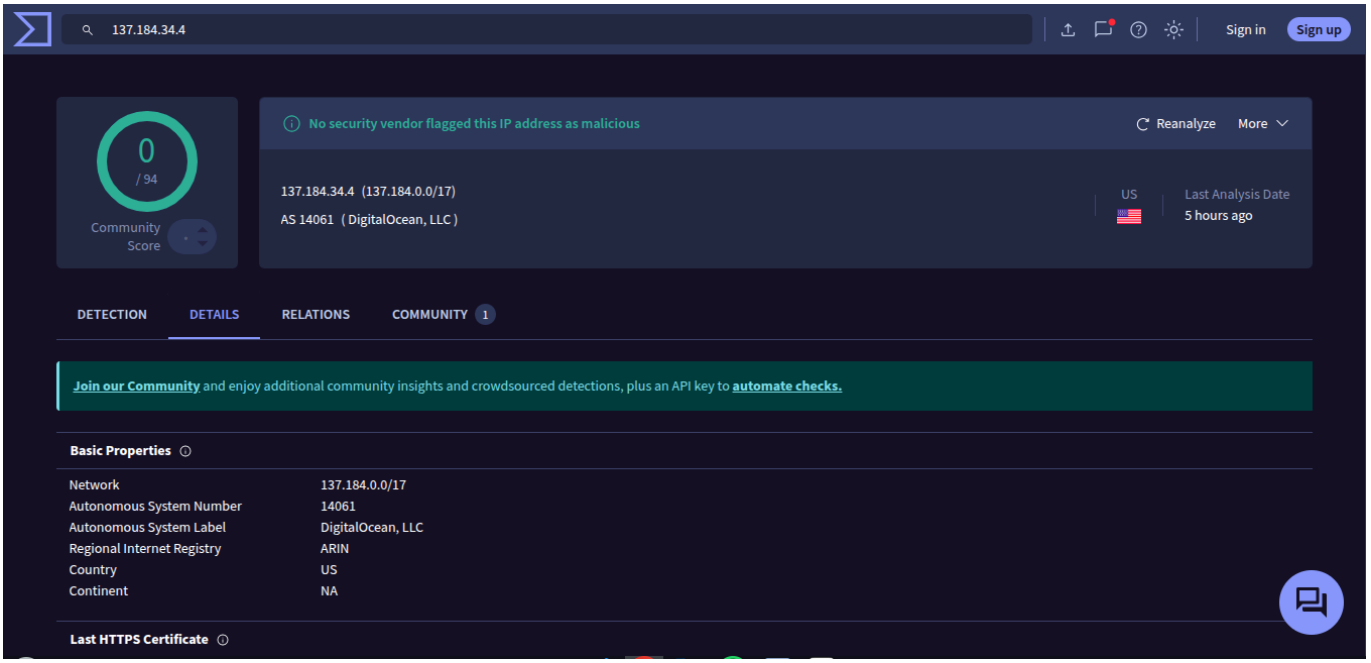
Once you have the originating IP address from the email header, you can investigate further to determine where the email actually originated.

1. Note the x-originating-ip address from your MX Toolbox analysis.

```
the email was not sent from a web ui/ by a user so there is no x-
originating-ip, in this case the x-sender-ip can be concidered as one
which is : 137.184.34.4
```

2. Use the originating IP in VirusTotal, AbusePDB, and/or Cisco Talos

analysis website



3. Record:

Field	Result
Detection counts	0
Reputation labels	1:not reccomended, 56:clean, 37:unrated
ASN/hosting provider	AS14061 - DigitalOcean, LLC
Any abuse history	Yes, 10 reports

4. State whether the IP ownership is compatible with a legitimate Bradesco/Livelo campaign.

```
No : the ip ownership is not compatible with a legitimate Bradesco/Livelo
campaign
    This is because of the results found in ip analysis
        ip address : 137.184.34.4
        Type : vps instance
        ASN: DigitalOcean AS14061
```

5. Paste a screenshot of the IP lookup results below and submit

137.184.34.4 was found in our database!

This IP was reported **10** times. Confidence of Abuse is **0%**: ?

0%

ISP	DigitalOcean, LLC
Usage Type	Data Center/Web Hosting/Transit
ASN	AS14061
Domain Name	digitalocean.com
Country	United States of America
City	Santa Clara, California

Reporter	IoA Timestamp (UTC)	Comment	Categories
Study Bitcoin 🤗	2025-01-19 19:26:45 (1 year ago)	Port probe to tcp/13000 [srv125]	Port Scan
MPL	2025-01-19 19:07:53 (1 year ago)	tcp/8088	Port Scan
Lia	2025-01-19 18:47:52 (1 year ago)	[UFW] 8090 (TCP)	Port Scan
diego	2025-01-19 18:25:57 (1 year ago)	Events: TCP SYN Discovery or Flooding, Seen 3 times in the last 10800 seconds	DDoS Attack
diego	2025-01-19 04:40:02 (1 year ago)	Events: TCP SYN Discovery or Flooding, Seen 5 times in the last 10800 seconds	DDoS Attack
trentwiles.com	2025-01-16 02:15:36 (1 year ago)	Unauthorized connection attempt detected from IP address 137.184.34.4 to port 8090 [EWR]	Port Scan, Hacking
MPL	2025-01-16 02:06:11 (1 year ago)	tcp/8081	Port Scan
JPPO	2025-01-16 01:28:14 (1 year ago)	Hit port : 8001	Port Scan
diego	2025-01-16 01:17:09 (1 year ago)	Events: TCP SYN Discovery or Flooding, Seen 3 times in the last 10800 seconds	DDoS Attack
RAP	2024-12-14 12:17:52 (1 year ago)	2024-12-14 12:17:52 UTC Unauthorized activity to TCP port 8443. Web App	Port Scan, Web App Attack

Summary

ASN	AS14061 - DigitalOcean, LLC
Hostname	No Hostname
Range	137.184.32.0/20
Company	DigitalOcean, LLC
Hosted domains	0
Privacy	True
Anycast	False
ASN type	Hosting
Abuse contact	abuse@digitalocean.com

Task 5 - Domain ownership and URL/Domain analysis

1. Run WHOIS for:
    - atendimento.com.br
    - blog1seguimentmydomaine2bra.me
2. Record:

`atendimento.com.br`

Field	Result
Registrar	Under brazilian internet system Centralized (Registro.br only)
Creation date	20-Sep-2018
Expiry date	20-Sep-2027
Available ownership clues	domain is generic, not brand-owned; no explicit organizational attribution visible

---

`blog1seguimentmydomaine2bra.me`

This domain is not registered

Field	Result
Registrar	NA
Creation date	NA

Field	Result
Expiry date	NA
Available ownership clues	NA

3. Submit the extracted URL to urlscan.io and/or VirusTotal URL analysis. Record whether the URL:

`atendimento.com.br`

Field	Result
Redirects	No significant redirect observed
Resolves	Yes
Is already known as malicious or suspicious	No detection on virus total
The landing domain matches the claimed brand	NA : no specific brancd claim

`blog1seguimentmydomaine2bra.me`

Field	Result
Redirects	NA
Resolves	NA
Is already known as malicious or suspicious	NA
The landing domain matches the claimed brand	NA

Final Questions:

1. Why is a From vs Return-Path mismatch important in phishing analysis?

```
The From address is what the user sees, while the Return-Path indicates the
actual sending source used for bounce handling. A mismatch between these
fields tells sender spoofing, where the visible identity is forged to
appear trustworthy while the actual infrastructure is unrelated.

This is a key phishing indicator because legitimate organizations typically
align these fields or use controlled domains. A mismatch implies the
attacker is impersonating a trusted identity while sending from
unauthorized infrastructure.
```

2. What does dkim=none suggest in the context of this sample?

```
The result dkim=none indicates that the email was not cryptographically
signed using DKIM which tells us that there is no verification of domain
authenticity or message integrity.
```

3. How should an analyst treat SPF/ DMARC temperror compared with explicit fail or pass results?

```
A temperror indicates a temporary failure in performing the authentication
check, usually due to DNS lookup issues.
It does not confirm legitimacy like a pass, nor does it definitively
indicate spoofing like a fail.

However, from a security perspective, temperror should be treated as
suspicious or untrusted, especially when combined with other anomalies.
Analysts should not assume legitimacy and should rely on additional
evidence such as header inconsistencies and infrastructure analysis.
```

4. Which observations in this sample are strongest for enterprise blocking and threat-hunting?

```
The strongest indicators include:

- email's from vs return-path mismatch
- DKIM = none tells lack of authentication
- SPF/DMARC temperror tells authentication failure to validate
- Originating IP from generic VPS/cloud infrastructure
- Suspicious or non-corporate mail server hostname
- Email authentication result showing compauth=fail
- Spam confidence indicators

These signals are reliable for both automated blocking and threat-hunting,
as they indicate unauthenticated and potentially malicious email
infrastructure.
```